

7.1GHz–8.2GHz 3W Transmitter Module (VTX)

User Manual

V1.0

2026. 05



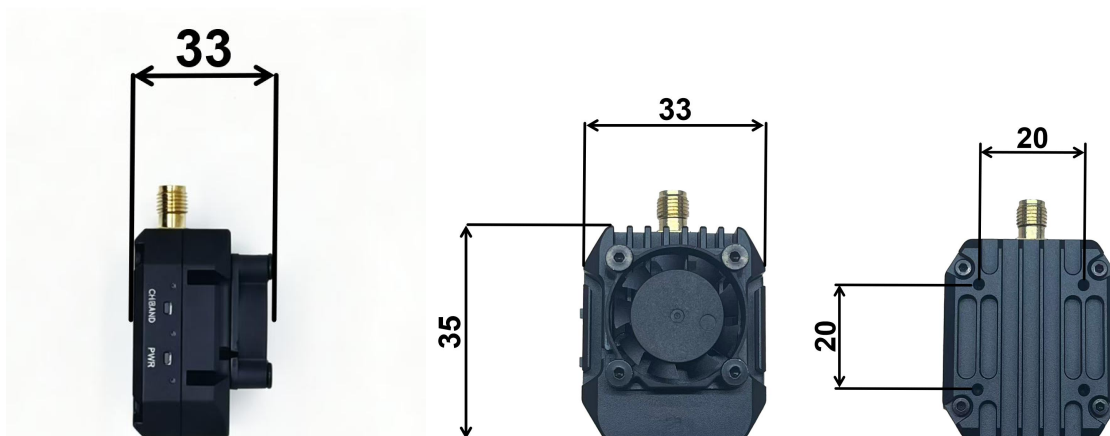
1.1 Part Name



1. SH1.0-8P cable
2. Channel Indicator (Red LED)
3. Band Indicator (Blue LED)
4. Power Indicator (Green LED)
5. Channel/ Band(CH/BAND) Button
6. Power Setting (PWR) Button
7. Antenna

1.2 Dimensions

The module dimensions and mounting dimensions are shown below (unit: mm); please reserve adequate installation clearance.



1.3 Specifications

- a. 64 channels: 7.1 GHz – 8.2 GHz (7110 MHz – 8210 MHz)
- b. Max. output power: 10 W (adjustable: 25 mW / 1 W / 2 W / 3 W)
- c. Supply voltage: DC 8 V – 36 V, 2 S–8 S Li-ion/Li-Po compatible
- d. Antenna connector: SMA female
- e. Control protocol: IRC Tramp
- f. Cooling: aluminum enclosure with heatsink and cooling fan
- g. Mounting pattern: 20 mm × 20 mm, $\varnothing$ 2.0 mm holes
- h. Weight: 56 g

2.1 Usage Instructions

2.1.1 Setting the Channel

Short-press the CH/BAND button to change the channel. Each press advances to the next channel; the corresponding red LED blinks once to indicate the selection (CH 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8, then wraps back to CH 1). See the diagram below for details.



Short-press the CH/BAND button to switch channels								
	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
RED LED	Blink once	Blinks twice	Blinks three times	Blinks four times	Blinks five times	Blinks six times	Blinks seven times	Blinks eight times

2.1.2 Setting the Band

Long-press the CH/BAND button to switch bands. Hold for 5 s: the blue LED blinks to indicate the selected band, cycling through groups A → B → E → F → R → P → H → U. Refer to the diagram below for details.



Long-press the CH/BAND button for 5s to switch bands								
	Band A	Band B	Band E	Band F	Band R	Band P	Band H	Band U
Blue LED	Blink once	Blinks twice	Blinks three times	Blinks four times	Blinks five times	Blinks six times	Blinks seven times	Blinks eight times

2.1.3 Setting Transmit Power

Press the PWR button briefly to cycle power levels: 25 mW → 1 W → 2 W → 3 W.

The green LED indicates the selected level by blinking:

- 25 mW: blinks once
- 1 W: blinks twice
- 2 W: blinks three times
- 3 W: blinks four times

See the diagram below for details.



Long-press PWR for 3s to enter Pit mode; the green LED turns solid on.

Short-press PWR to cycle power levels					
	Pit Mode	25mW	1W	2W	3W
Green LED	solid on (Long-press PWR for 3s)	Blink once	Blinks twice	Blinks Three times	Blinks four times

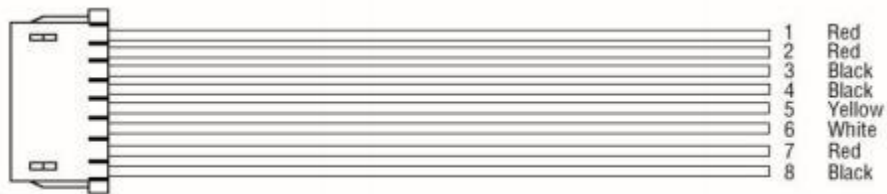
Note: This video transmitter (VTX) is equipped with thermal protection.

If the internal temperature exceeds 100 °C, output power is reduced by one step; if the temperature remains above 100 °C, it is reduced again, repeating until the minimum level (1 W) is reached. Once the temperature drops to 95 °C or below, the VTX automatically restores the originally selected power level.

2.2 Channel-to-Frequency Table

Channel-to-Frequency Table								
Band	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
Band A	7110	7130	7150	7170	7190	7210	7230	7250
Band B	7270	7290	7310	7330	7350	7370	7390	7410
Band E	7430	7450	7470	7490	7510	7530	7550	7570
Band F	7590	7610	7630	7650	7670	7690	7710	7730
Band R	7750	7770	7790	7810	7830	7850	7870	7890
Band P	7910	7930	7950	7970	7990	8010	8030	8050
Band H	8070	8090	8110	8130	8150	8170	8190	8210
Band U	7115	7265	7415	7565	7715	7865	8015	8165

2.3 SH1.0-8P Pinout Description



Pinout Description

NO	1	2	3	4	5	6	7	8
FUN	18-36V		GND		Video	IRC	5V Out	GND
Color	Red		Black		Yellow	White	Red	Black

3 Precautions

3.1 When installing the VTX, leave adequate space around the module to guarantee air-flow and proper heat dissipation. Otherwise the over-temperature protection will activate, reducing—or even shutting off—transmit power.

3.2 Before powering on, verify that the supply voltage is within the specified range and that polarity is correct to prevent damage to the unit.

3.3 Always connect an antenna to the RF output before applying power; this extends the module's service life.

3.4 Read this manual before use to ensure correct wiring and prolong the life of the module.

3.5 Do not bend the antenna cable or power leads sharply; improper handling can degrade performance or cause failure.

3.6 Understand and comply with all local laws and regulations before operating.

3.7 This product is not intended for use by children.